

BODAPATI VISHNU SAI VARDHAN

Game Developer | XR Developer | Spatial Computing Builder | Technical Creative

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PROFILE

Game and XR developer pursuing an M.Eng in Game Design, Development and Innovation at Duke University, with hands-on experience building Meta Quest mixed reality tools, clinical VR rehabilitation interactions, Unity/Unreal game systems, and AI-enabled web products. Strong focus on spatial interaction, gameplay systems, performance-aware XR implementation, UI clarity, and rapid prototyping from concept to playable demo.

EDUCATION

Duke University - Master of Engineering, Game Design, Development and Innovation

Aug 2025 - Present

Pratt School of Engineering | Key coursework: Advanced Game Design, Unreal Engine Development, C++ Programming

University of Wollongong - Bachelor of Computer Science, Game and Mobile Development

Sep 2021 - Sep 2024

Key coursework: Game Engine Essentials, VR/AR Game Development, Interactive Computer Graphics, Mobile Application Development, IT Project Management

AWARDS & HACKATHONS

1st Place - Grand Winner, DesignXR Hackathon 2026

MR Blueprint - Mixed Reality Physics Sandbox

- Awarded Grand Winner for a Unity-based Meta Quest mixed reality sandbox enabling interactive 3D scene creation, drawing, editing, and physics simulation.

Winner - Honorable Mentions, Team USA x Google Cloud Hackathon

Draft USA - The LA28 Momentum League | Devpost project | [Project Link](#)

- Built a Gemini-powered fan engagement app for drafting Team USA sports, Olympic-Paralympic pairings, and hometown hubs through transparent momentum scoring and AI-assisted roster guidance.

Top 50 Semifinalist - DevStudio 2026 by Logitech Hackathon

MR Blueprint - selected from a hackathon with 1,323 participants

- Advanced to the Top 50 semifinalist stage for a spatial computing prototype using Meta Quest, Unity XR, and Logitech MX Ink stylus interactions.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant - Duke Intelligent Interactive Internet of Things (I3T) Lab

Oct 2025 - Present

VR Gameplay & Research Developer | Duke University

- Develop VR-based rehabilitation gameplay for ICU patients, prioritizing seated/fixed-position comfort, accessibility, clear feedback, and constrained patient movement scenarios.
- Implement and iterate on Unity/Meta Quest gameplay mechanics, tutorial and calibration flows, interaction feedback, and repeated-session reliability for clinical research use.
- Optimize frame-rate stability, input responsiveness, and interaction smoothness to reduce motion discomfort risk in clinical VR settings.
- Collaborate with researchers and developers to translate clinical requirements into usable gameplay systems, testing workflows, and patient-friendly interaction design.
- Evaluate integration paths for Gaussian Splatting and AI-generated scenes inside Lumi to support richer, more realistic therapy environments.

Freelance Research Analyst - Singapore Gaming & Graphics Startup Ecosystem

Oct-Dec 2024

Exelvision IT Labs LLP

- Conducted market research across gaming, graphics, robotics, 3D modelling, VR/AR development, and artificial intelligence sectors.
- Compiled industry trends, startup performance metrics, investment activity, competitor strategies, and emerging technology opportunities for client-facing reports.
- Produced data-driven insights that supported strategic decisions, business development initiatives, and sector-specific opportunity evaluation.

SELECTED PROJECTS

MR Blueprint - Mixed Reality Physics Sandbox

Jan-Apr 2026

Unity | C# | Meta Quest 3/3S | OpenXR | XR Interaction Toolkit | Meta XR SDK | Logitech MX Ink SDK

- Built a Meta Quest mixed reality physics sandbox where users can spawn, edit, draw, and simulate interactive 3D physics scenes inside an XR workspace.
- Implemented core C# systems for object spawning, selection, transform manipulation, world-space XR UI, inspector panels, and Edit/Simulate Mode physics configuration.
- Contributed Logitech MX Ink Draw Mode and PhysicsLens features, including spatial drawing, pressure-sensitive strokes, live graphing, physics visualization, and snapshot/restore workflows.

SELECTED PROJECTS (CONTINUED)

Draft USA - AI Sports Draft Strategy Platform

May 2026

Next.js 14 | TypeScript | Tailwind CSS | Gemini API | Google Cloud Run | Docker | Recharts | [Devpost](#)

- Built a responsible AI fan engagement app that turns Team USA sport discovery into a fantasy-style draft experience focused on momentum, parity, hometown breadth, and underfollowed sports.
- Integrated Gemini-powered features for scout recommendations, roster analysis, parity coaching, leaderboard explanations, and Commissioner Recaps using server-side API routes.
- Designed user-facing dashboard flows for Draft Room, Roster Dashboard, Hometown Hubs, Momentum Score breakdowns, bonus unlocks, and league storytelling.
- Implemented responsible-language constraints that frame scores as fan discovery signals rather than medal predictions, betting signals, or athlete rankings.

Tower of Tricks - UI Programmer | Build & Debugging

Aug-Dec 2025

Unreal Engine | Blueprints | UMG | Packaging | QA

- Designed and implemented menus, HUD elements, settings/pause flows, and player-facing feedback systems for a playable Unreal game project.
- Debugged editor-vs-packaged build issues related to level loading, map inclusion, asset references, initialization order, and gameplay stability.
- Managed packaging configuration and performed systematic QA across builds to validate gameplay, UI behavior, and deployment readiness.

Hungry Owl - Sole Developer

Aug-Dec 2025

Playdate | Lua | Gameplay Systems | UI

- Designed and developed a complete Playdate game from scratch, owning gameplay, systems, UI, scoring, enemy behavior, difficulty progression, and game state logic.
- Integrated hardware-specific input using the Playdate crank and buttons, mapping physical controls to responsive in-game aiming and interaction systems.

Overpriced - Game Jam

Nov 2025

Gameplay Programmer | Audio Programmer | Unreal Engine

- Implemented the core audio integration pipeline, including sound cue setup, event triggering, and contextual audio feedback tied to gameplay interactions.
- Programmed gameplay interaction logic supporting player-environment connectivity, puzzle flow, level progression, and responsive feedback states.

The Misadventures of Skylar Knight - Game Jam

Jan 2026

Gameplay Programmer | Audio Engineer | UI Developer | Unreal Engine

- Implemented player mechanics, interactions, core game logic, UI/audio feedback, 3D-to-2D gameplay adaptation, build testing, packaging, and version control support.

Meteor Mayhem - Hand Gesture-Based Game

Python | Computer Vision | Unity | C#

- Developed a gesture-controlled Unity game using Python-based real-time hand gesture recognition to drive in-game actions and natural interaction flow.

Additional Unity Academic Prototypes

FPS prototype | VR Maze | Interactable VR Room

- Built Unity prototypes covering first-person mechanics, enemy/weapon systems, VR locomotion, spatial audio, haptic feedback, object interaction, and physics-based manipulation.

TECHNICAL SKILLS

Engines & XR: Unity, Unreal Engine, Meta Quest 3/3S, XR Interaction Toolkit, OpenXR, Meta XR SDK, Logitech MX Ink SDK, URP, world-space XR UI, interaction design

Programming: C#, C++, C, Python, JavaScript, TypeScript, Java, Lua, SQL/MySQL

Gameplay & Systems: Gameplay programming, physics interactions, UI/HUD systems, audio implementation, game states, input systems, build packaging, QA/debugging, performance optimization

Web, Cloud & AI: React, Next.js, Tailwind CSS, Gemini API, Google Cloud Run, Docker, server-side API routes, prompt/agent constraints, responsible AI design

Tools: Git/GitHub, Android Studio, VS Code, Netlify, Formspree, basic video/editing workflows, documentation, technical research

CERTIFICATIONS, LEADERSHIP & LANGUAGES

Certifications: Young AI Pro Training Program - Advanced AI & LLM Techniques (2024); Game Development With Unity (2020, MAGES Institute of Excellence); Professional Remote Pilot License Program (2019, Ace Aviation); UI Design Course (2020, Udemy); User Experience (UX): The Ultimate Guide to Usability and UX (2020, Udemy)

Leadership: Entrepreneur Society Captain - led fundraising and supported event organization (2018)

Languages: English (fluent), Telugu, Hindi

Updated CV - July 2026